

CAmkES tutorials bug fixes

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hello-camkes-0 Tutorial

When trying to build and run the tutorial from <https://docs.sel4.systems/Tutorials/hello-camkes-0.html>, with

```
# In build directory
ninja
```

the process gets stuck without an error message.

- **Expected behavior:** after the ninja command it should have successfully built the hello-camkes-0 tutorial
- **Real behavior:** The process gets stuck running the "ninja" command without displaying any error message.
- **Cause:** The exact cause did not become clear during bugfixing. It is possible that specifying the "--platform" parameter in the command ". /init --tut hello-world" (as shown in the example code) could have led to this behavior.
- **Fix:** What helped was to specify the platform with the "--platform pc99" directly (even though this should already be the default value).



This issue seems to be more general as I have faced it in the [sel4 threads tutorial](#). Selecting the pc99 platform seems to fix the problem.

In my case it code stuck on the following step:

```
in-container:/host$ ./init --tut threads...in-container:/host$ cd threads_build/in-container:/host
/threads_build$ ninja...
[181/189] Generating capDL-tool/parse-capDL
```

Timer Tutorial

Unfortunately, the timer tutorial (<https://docs.sel4.systems/Tutorials/hello-camkes-timer.html>) has a bug:

- **Expected behavior:** after the main thread starts the program should print

```
-----Sleep for X seconds-----
```

and with a delay of x seconds a second message

```
After the client: wakeup
```

- **Real behavior:** the program prints out the 2 messages without a delay.
- **Cause:**
 - `ttc_start()` starts the timer and it immediately generates an interrupt
 - in the `irq_handle()` we release the semaphore and stop the timer
 - the timer will not generate interrupts anymore and the semaphore is not locked so the delay in the hello sleep will not occur. `hello_sleep()` is executed after the `hello_init()` and `irq_handle()`
- **Fix:**
 - remove the `ttc_start()` function from `hello_init()`
 - move it to `hello_sleep()` after the `ttc_set_timeout()` (we first define the delay and then start the timer)
 - `ttc_stop()` in `irq_handle()` is not necessary because we define the timeout as not periodic in, as the last argument passed to `ttc_set_timeout(ttc_t *ttc, uint64_t ns, bool periodic)` is false. But it will not break the functionality